CHAPTER 14

Deploying to the Cloud

The cloud: our final frontier. We've already seen how to deploy to a local Swarm cluster (albeit with only a single node). In this chapter, we take this a step further and create infrastructure running in the cloud. We'll start by creating a cluster on DigitalOcean, and then see how similar this is to creating a cluster on Amazon Web Services (AWS).

By the end of this chapter, we'll have our application deployed and running in both DigitalOcean and AWS. Although our cloud deployment won't be production-grade, you'll have gained some valuable experience and started to get a feel for some of the real-world considerations at play.

Note that, as the cloud services we use in this chapter aren't free, some (small) costs are associated with running the application. However, assuming you're diligent and clean up or stop the resources when we're done, the costs will be neglible—measured in the pennies.

Creating a DigitalOcean Cluster

Before we can deploy our app to the cloud, we need cloud-based instances on which to deploy it. We're going to start by creating a Docker swarm on DigitalOcean. The prices[1] are very modest: $0.007 per hour for the smallest, standard *Droplet*—their term for an instance. In real terms, a three-node cluster running for 48 hours will cost $1.

To follow along, you'll need your own DigitalOcean account. Go to the DigitalOcean homepage and complete the sign-up form.[2] You'll need to enter your credit card details, since there is a charge for using their cloud resources.

1. https://www.digitalocean.com/pricing/
2. https://www.digitalocean.com